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Revision	Description	Date
A	Initial Release – (John Tedesco)	(12/4/09)

Subject: (Detector Signals Input Test with a SAM Module)

System(s) Affected: (Pre-NS5 Dimension Systems)

Action Category: (As Required)

Summary: (It is often difficult to determine what parts are causing detector signals to fail. This will help in the search by eliminating some of the possibilities.)

Tools Needed:

-Voltmeter

-Signal Access Module

1. Place the signal access module between the NS4 controller and microscope or between any extender and the microscope. Make sure all switches are set to output
2. Place the SW in realtime and have the nanoscope controller on
3. The BNC connectors labeled as below will have the deflection and sum signals that we are looking for

AUXA – Lateral (Horizontal) Deflection

AUXB – Vertical Deflection

AUXC – Sum Signal

4. Place your laser on your cantilever and center the spot in the oval viewing window on your scanner
5. Use a voltmeter (make sure to set for DC voltage) to test each voltage (+/-15V and +5V)



The outside of each BNC connector can be used as the reference ground (touch the black lead to it). The inside part of the connector is the "hot" portion. Use the red lead there. If all voltages are not present do not move on to the next step.

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6. Next, start with AUXC and measure the Sum with the voltmeter. It should be a negative voltage. For tapping mode tips with no aluminum coating this value should be near -2V, for aluminum coated tips near -4V, for SiN gold coated tips near -6V.
7. Remove the laser spot from the tip by adjusting laser position and see if AUXC goes to near 0V. Put the laser back on the tip and get a good spot in the center of the oval viewing window
8. Next test the vertical deflection (AUXB). When testing use the vertical deflection adjust knob (pic in appendix A) on the left side of the scanner. When rotating this knob it should be possible to get this pin to go to +/-12V. for example, hold the voltmeter's red lead to the inside of the AUXB connector, hold the black lead to the outside of the AUXB connector and then rotate the upper left knob of the dimension scanner all at the same time (yes, you will probably need 2 people). The values should change as you move the knob. Counter-clockwise should move the voltage more negative, clockwise more positive.
9. Do the same thing as in step #11 but this time on AUXC and rotate the bottom left knob of the scanner to adjust horizontal deflection (see appendix A for location of knob).
10. If all values come through properly on the test pins yet the detector screen is not reading them properly then the problem not in the scanner or microscope and somewhere in the extender and controller. If the values do not come through as expected then the scanner or the electronics in the Dimension are to blame.

Appendix A:

Vertical Deflection
adjust knob

Horizontal Deflection
adjust knob

